Product Semantics in Theory and Practice

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Abstract

In the field of industrial design it is important that products, through their material form, be given an opportunity to carry and communicate an intentional message. A product can express characteristics and thereby create desirable emotions in a potential user.
Semiotics is usually described as sign theory – the science and study of signs and sign systems, their structure, properties and role in sociocultural behaviour. The term 'sign' has been defined thus: *'A sign, or representamen, is something which stands to somebody for something in some respect or capacity.'* (Charles S. Peirce, CP 2.228.) Semiotics may be seen as a way of observing with which people's perceptions and attitude toward the surroundings can be described and explained. A sign is something that conveys a message to us. *Semantics* is the theory of relationships between signs and their meanings, and studies the product's message.
The *icon* is a category of signs that builds upon similarity between physical form and what it refers to, such as another product, an activity, or a property or expression. Icons occur commonly in design work.
Metaphors can be regarded as icons, and a metaphor may be used to give a product some specific expression. The product may express characteristics that resemble those of another product, and their relationship can then be called metaphorical. A sports car expressing speed, manliness and efficiency may be used as a metaphor for creating these expressions in, for example, a vacuum cleaner.
Product semiotics is fundamental to the instruction given today in the educational area of Industrial Design Engineering (IDE), among others, at Chalmers University of Technology. Semiotic concepts and models form a basis for a product-semiotic theory and, with this point of departure, the *product sign* can be analysed and explained. The instruction contains a theoretical part and a more practical applied part. Through theoretical instruction in product semiotics, the students gain access to a *conceptual apparatus* which makes it possible to *analyse* and discuss an existing product's semantic functions and to *formulate requirements* for a new product's semantic functions. Through partly experimental exercises and application projects, the students proceed to formulate, using specified semantic requirements, a new product solution with a focus on the product's semantic design.
In one application project, the aim was that the students should free themselves completely from the current product sign and think of entirely new design solutions for products – a camera, a radio, and an alarm clock. The starting point of the project was to consider the product as a sign and to define a desired message as the basis for the semantic design. The students began with a specific life-style and formulated a number of *expressions* that were to attract the envisaged user. The life-style was illustrated with an 'image board', and the chosen expression with an 'expression board'. The latter serves to visualise a particular expression by means of a metaphor, a (natural or product) form, a colour, a material or surface structure, and an existing product that can be said to carry the expression. Subsequently, the 'expression board' provides a basis for sketch-work in which the final product is given a form that carries the specified expression.

Through the experiences acquired during the instruction, it has been shown that product semiotics is applicable in practical design work. This paper will illustrate two examples that clearly demonstrate how a specified expression can be shaped through the product.

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Introduction

The word *design* comes originally from the Latin *designare* which means to mark out, designate, denote[1]. In the field of industrial design it is important that products, through their material form, be given an opportunity to carry and communicate an intentional message. The designer communicates by means of the product. What we see, hear and feel of a product tells us something. A product can express characteristics and thereby create desirable emotions in a potential user.

This paper describes a product semiotic theory and shows how one can practically apply the theory in design work, where the designer proceeds from a specific life-style and a specified expression, and consciously charges the product with a message. The paper will illustrate two examples that clearly demonstrate how a specified expression can be shaped through the product.

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Theory

Semiotics

People use signs in many different contexts, and semiotic theories have been applied within several fields of research – especially in cultural and linguistic sciences, but also in other areas such as art, architecture, music and film (Nöth 1990). Semiotics originated in antiquity, and the philosopher Plato established the first ideas surrounding the concept of signs. Within semiotics, models and concepts described by Ferdinand de Saussure (1857 – 1913), Charles Sanders Peirce (1839 – 1914) and Charles William Morris (1901 – 1979) play a leading role and form the basis for modern semiotic theory.

Design semiotics has been developed from semiotics and provided design with a usable theory and terminology, where also semioticians such as Karl Bühler, Roland Barthes, A. J. Greimas, Umberto Eco and Yuri Lotman have become important (compare Vihma (ed.) 1990, 1992, Väkeva (ed.) 1990, Tahkokallo & Vihma (eds.) 1994, Vihma 1995, Monö 1997). Semiotics is usually described as sign theory – the science and study of signs and sign systems, their structure, properties and role in sociocultural behaviour. The term 'sign' has been defined thus: *"A sign, or representamen, is something which stands to somebody for something in some respect or capacity."* (Charles S. Peirce, CP 2.228.) Semiotics may be seen as a way of observing with which people's perceptions and attitude toward the surroundings can be described and explained.

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Signs

Charles S. Peirce sees the sign as a kind of triad – *representamen (R), object (O) and interpretant (I)* – in which each unit has a close relationship to each of the others (Figure 1). The form taken by the sign functions as a sign – *representamen* – and refers to something apart from itself, which is what the sign stands for – the *object* – and is understood or interpreted by someone, that is, it acquires a meaning or implication in the receiver's consciousness – the *interpretant.* The interpretant is not the sign's user, but what Peirce calls "the actual significant effect", i.e. the *mental conception* that is created by both the represented form and the user's experience of the object, as a consequence of the interpretation. The individual interprets the relationship R-O. This relationship then forms semantic consequences.

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[1] Etymological Dictionary of the English Language, Oxford: Clarendon Press, 1988.

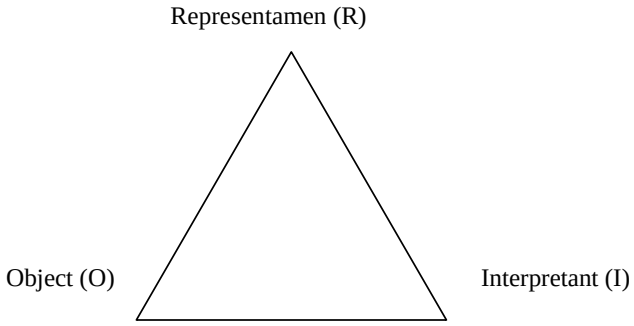


Figure 1: Peirce's sign triad

The sign is a factor in a process of communication where both the sender and receiver play an active role. In all cultures, communication processes occur which are classified into many languages (verbal, visual etc.). Semiotics is a means of studying these communication processes in the form of signs and sign processes. The relationship R-O-I is a sign process in which the three parts are connected with each other and together create the sign. The term that is used to describe the sign process is the concept of *semiosis* (Nöth 1990). Semiosis can be regarded as *unlimited*, i.e. a sign may actualise a new or different sign in the interpreter. This means that a sign/representamen can refer to another sign/object which is stored in the receiver's memory and then functions as a representamen in the next semiosis. In Peirce's triad, the sign's content is focused by his definition of the two units 'object' and 'interpretant'. This renders the triad usable in design research (Wikström 2002). Knowledge about the relationship between the three units in the triad is important for a product analysis and when formulating semantic design, i.e. the product's form and message (cf. Vihma 1995).

In design semiotics, Vihma (1995) has specifically studied and applied Peirce's sign model for products. Vihma assumes that the representamen is a visual construction that can refer to the object. The representamen may be a physical thing, a word for the thing, or a mental picture of the thing. The object may be another thing, activity, function, event, quality, or property. According to Vihma, the physical product can be manifested in the object, in the representamen, or in the relationship between these two parts of the triad, due to the very unlimitability of semiosis.

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Icon, index and symbol

Peirce divides signs (Figure 2) into three *sign categories – icon, index and symbol* – where each category displays a different relationship between the representamen (R) and the object (O), that is, how R refers to O (Nöth 1990).

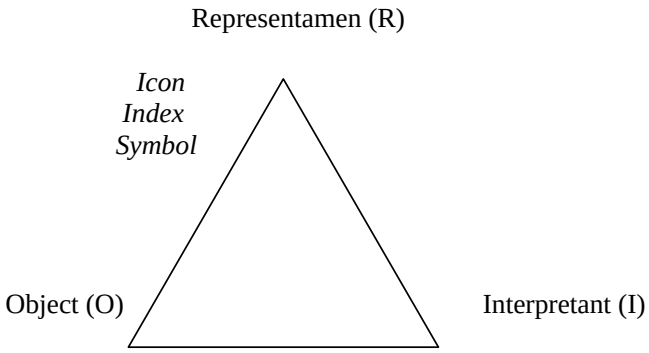


Figure 2: Peirce's sign triad, Icon, Index, Symbol

The *icon* is a category of signs that builds upon similarity between physical form and what it refers to, such as another product, an activity, or a property or expression. Icons occur commonly in design work. *Metaphors* can be regarded as icons (Vihma 1995), and a metaphor may be used to give a product some specific expression. The product may express characteristics that resemble those of another product, and their relationship can then be called metaphorical. A sports car expressing speed, manliness and efficiency may be used as a metaphor for creating these expressions in, for example, a vacuum cleaner. Metaphors are ubiquitous and have long been applied practically in the design of products, without knowledge or awareness of semiotics. What semiotics contributes to design work is that these aspects can more systematically be discussed and used as a means to formulate an intended message, and also to identify a false or undesirable message, in the product (Wikström 2002).

The *index* builds upon a closeness or a causal connection between the representamen and the object. An indexal sign has a causal relation to something other than itself. Such an example is a sign consisting of an arrow which refers to a definite direction for evacuating a locality. A represented form may refer to usage (Vihma 1995); for instance, a doorknob can describe the way it should be gripped and turned. Sound is a case of an indexal sign: a 'rattling sound' when one lays a product on a table, or a pleasant 'click sound' from a car door (the Mercedes click), can give us an indication of quality and finish in the product.

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A *symbol* means something because of an agreement between people. There is no causal connection between the representamen and the object. The symbol is a mental and social construction that builds upon learning. If the symbol does not exist in our memory, it is incomprehensible to us. The individual must have knowledge of both the symbol's meaning and the context in which it is used, so as to acquire an understanding of its significance. Examples of symbols are various types of graphic signs. Graphic symbols are used for different functions on buttons and controls – such as a symbol for the function on/off that electronic products have.

Vihma (1995), in a dissertation, has focused on the relationship between the representamen and the object, and studied how Peirce's three *sign categories – symbol, icon and index* – are used for formulation and analysis in design theory. These categories require different knowledge in the user, so that the sign can be interpreted correctly.

In a product's graphic design and the choice of sign category, the designer should be well aware of, and acquainted with, the problems that can arise in the situation of use. For instance, there may be problems with handling controls, which can occur if one uses a *symbol* rather than an *icon* as the bearer of specific content for a task or activity (Wikström 2002).
Symbols should be avoided entirely for products that are to serve in a public environment, for example is the present design of stamping automats in Gothenburg. For a trip, the traveller is to pay by stamping a magnetic card in an automat, which has ten buttons marked with the numbers 1 – 10. The traveller must choose one of the buttons so that the right sum is drawn from the card. If, say, an adult is travelling within the central parts of the city, *the button with the number '2'* should be chosen. Based on an analysis of *the product as sign*, this button is a *symbol* for *'an adult person who travels within a certain zone in Gothenburg'*. As mentioned earlier, the concept of a 'symbol' is defined as an agreement between people, which means that first-time users plausibly cannot understand its meaning directly without additional information, such as text or help from the driver or other passengers.

Product semantics

Product semantics has been defined as *"a study of the symbolic qualities of man-made forms in the cognitive and social context of their use and application of knowledge gained to objects of industrial design."* (Krippendorff & Butter 1984.) Thus, product semantics concerns the relationship between, on the one hand, the user and the product and, on the other, the importance that artefacts assume in an operational and social context.
A sign is something that conveys a message to us. *Semantics* is the theory of relationships between signs and their meanings, and studies the product's message. Thus we can say that *"a sign is any phenomenon which has a significance that is independent in relation to its material form" (Monö 1997)*. In other words, a sign is something from which a meaning is generated. According to this definition, the *product gestalt* can obviously be a sign, and semiotics can be applied in design theory. Within design semiotics, a thing is focused upon as a sign-bearer, along with its ability and possibilities to bear and convey a message.
In the field of industrial design, it is important that the product, through its material form, is given an opportunity to bear and convey an intended message to a user. If we cannot understand the product's meaning, its function, through its design, it becomes incomprehensible to us and can thereby have negative consequences in the situation of use.
Monö (1997) has chosen, on the basis of theories of product semantics, to describe the product as a trinity. The first dimension, the *ergonomic* whole, includes everything that concerns the adjustment of the design to human physique and behaviour when using the product. The *technical* whole stands for the technical function of the product, its construction and production. The third aspect, the *communicative* whole, refers to the product's ability to communicate with users and its adjustments to human perception and intellect. Through the product gestalt, i.e. the totality of colour, material, surface, structure, taste, sound etc. appearing and functioning as a whole, the product communicates a message which is received and interpreted by the customer/user. This message, according to Monö (1997), is 'created' by four semantic functions, which are:

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- *To identify.* The product gestalt identifies e.g. origin and product area. A bowl can be identified as a part of a specific china set; a company can be identified by its trademark, or by a specific design philosophy apparent in its products.
- *To describe.* The product gestalt can describe the product's purpose and function. It can also describe how the product should be used and handled. For instance, a doorknob can describe how it should be gripped and turned.
- *To express.* The product gestalt expresses the product's properties, for instance 'stability', 'lightness' or 'softness'.
- *To exhort.* The product gestalt triggers the user to react in a specific way without contemplating or interpreting the product's message. For instance, the user is triggered to be careful to be precise in his/her operation of the product.

The product examples described in this paper, however, focus on the function 'to express'.

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The product's attractiveness

Products, through their message and semantic functions, should arouse feelings and recognition and action. The user should be able to form a perception of the product's properties – its *attractiveness* – in such a way that he/she wants to use the product (Figure 3).

Aesthetic attractiveness is part of the semantic message, namely what the product is to express. This in turn should be anchored in the company's business idea and the market on which the product is to be launched. An object can be attractive in many different ways and on numerous grounds, not least as consequences of social values and life-styles. The product's attractiveness may in certain cases be only an immaterial aspect. The individual may have an inner experience when he/she sees the product, or a private association – a message that need not be formulated by the designer. Carl Erik Linn (1985) employs the term *meta-product* for the commercial product's immaterial aspect. All the associations that the product or trademark gives rise to are components of the meta-product.
Knowledge of product semiotics should be an important tool for the product-developing company when creating new products. To have competence and a comprehensive picture of the user's requirements regarding the product's *communicative aspects*, the company must take account of both the product's cognitive and emotional qualities (Karlsson & Wikström 1999, March 1994). To achieve this, semiotics can offer concepts that are applicable and *enable a dialogue between the customer/user, the designer and the client/company, or between different departments within the company* (such as the method/testing department and the product development department) concerning the aspects that constitute the product's communicative unity. Hence it should be possible to clearly formulate the product's *"attractiveness"* in the form of semantic requirements for a product's expression, which can be directed toward a specific customer category or target group. These expressions then have semantic consequences in terms of recognition, understanding, emotions etc. (Figure 3.)

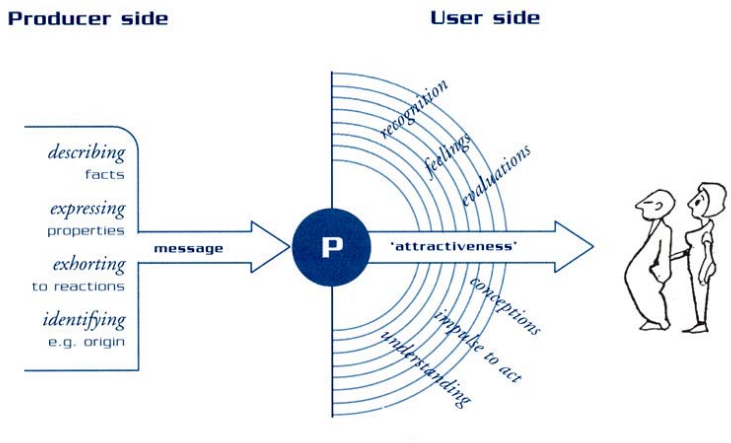


Figure 3: The product's four semantic functions and its consequences in terms of recognition, understanding, emotions etc. (Monö 1997)

Negative expressions

What is striven for in a good design is that the product's factual properties are clearly conveyed, i.e. that the product expresses the properties which it has. Equally important is that the product gestalt is not given a *false expression*, i.e. it must not express properties which it lacks.

From a user perspective, it is important that the product's *expression* agrees with other semantic functions, as well as with technical and ergonomic functions. If the product has an expression of, for instance, *security* and if this does not coincide with the product's factual properties, it can have disastrous consequences in the situation of use. The product then has a false expression that can *challenge* the user to act undesirably (Wikström 2002, Karlsson & Wikström 2006).

An expression can be evaluated as positive in a certain context while, in another context and for a different group of users, the same expression can receive a negative value. People interpret and evaluate the product's message in relation to their own experiences and to a particular context, in terms of acceptance or rejection, of liking or disliking. The individual identifies himself/herself, to varying degrees, with a role that the product and context are connected with. The product, through the semantic design, its expression, may either strengthen or weaken this role, thus creating positive or negative experiences, feelings, values and associations in the individual (Figure 3). For instance, a handicap aid may be designed so that it strengthens the individual's role of being handicapped, and the individual may then choose to refrain from using the product.

Applying theory

Product semiotics is fundamental to the instruction given today in the educational area of Industrial Design Engineering (IDE), among others, at Chalmers University of Technology. Semiotic concepts and models form a basis for a product-semiotic theory and, with this point of departure, the *product sign* can be analysed and explained. The instruction contains a theoretical part and a more practical applied part. Through theoretical instruction in product

semiotics, the students gain access to a *conceptual apparatus* which makes it possible to *analyse* and discuss an existing product's semantic functions and to *formulate requirements* for a new product's semantic functions. Through partly experimental exercises and application projects, the students proceed to formulate, using specified semantic requirements, a new product solution with a focus on the product's semantic design.

In one application project, the aim was that the students should free themselves completely from the current product sign and think of entirely new design solutions for products – a camera, a radio, and an alarm clock. The starting point of the project was to consider the product as a sign and to define a desired message (object in the sign triad) as the basis for the semantic design (representamen in the sign triad). The students began with a specific life-style and formulated a number of *expressions* that were to attract the envisaged user. In order to come closer and render the life-style more vivid, so-called 'storytelling' was employed. The student was given the task of imagining a specific person who represented that life-style and writing a short narrative about the person. The story also included a description of the person's relation to the intended product.

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The life-style was illustrated with an 'image board' (Figure 4a,b), and the chosen expression with an 'expression board' (Figure 5a,b). The latter serves to visualise a particular expression by means of *a metaphor, a form* (natural or product), *a colour, a material or surface structure, and an existing product* that can be said to carry the expression. Subsequently, the 'expression board' provides a basis for sketch-work in which the final product is given a form that carries the specified expression. The sketch-work began with 'thought traces' – or traces of thinking – on paper, so as to capture different forms which the metaphor, the product etc. lead to. With thought traces one can wonder, question, search, and test the expression's visual character before the final product form is created in Studio Tools (Figure 6a, b).

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Through the experiences acquired during the instruction, it has been shown that product semiotics is applicable in practical design work. Below are presented two of these examples, where the students themselves have formulated a life-style and an expression and, on this basis, worked with an 'image board' and an 'expression board'.

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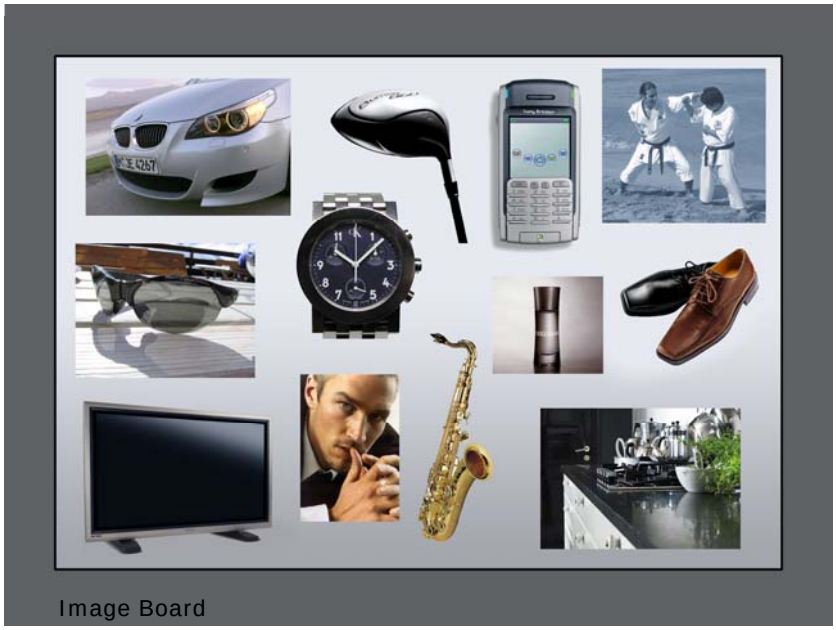


Figure 4a – Image Board

Example 1[2]

The Japanese company Canon has developed a new series of digital cameras which addresses the modern, successful male in his thirties. The model series will bear the name CRYSTAL. Its design is pure in style, with a point of departure in the expression 'stylistically pure'.

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Figure 5a – Expression Board – the expression 'stylistically pure'.

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[2] Design Lars Norin 2005.



Figure 6a – Model in Studio Tools

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Figure 4b – Image Board

Example 2[3]

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The intended target group is just over age 50 and enthusiastic about gardening. A radio for listening to while weeding or for coffee in the bower. The expression is verdant.

[3] Design Karin Lidman 2005.

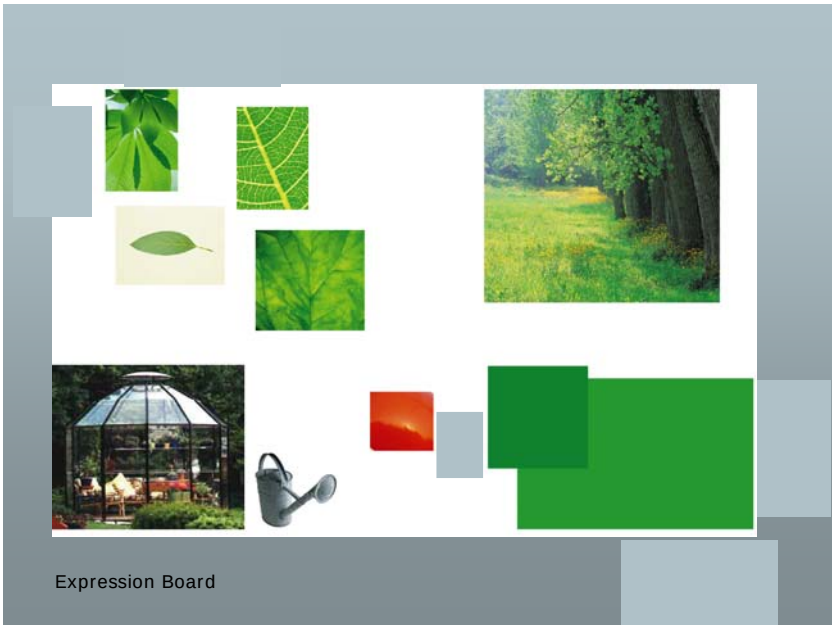


Figure 5b - Expression Board – the expression ‘verdant’.

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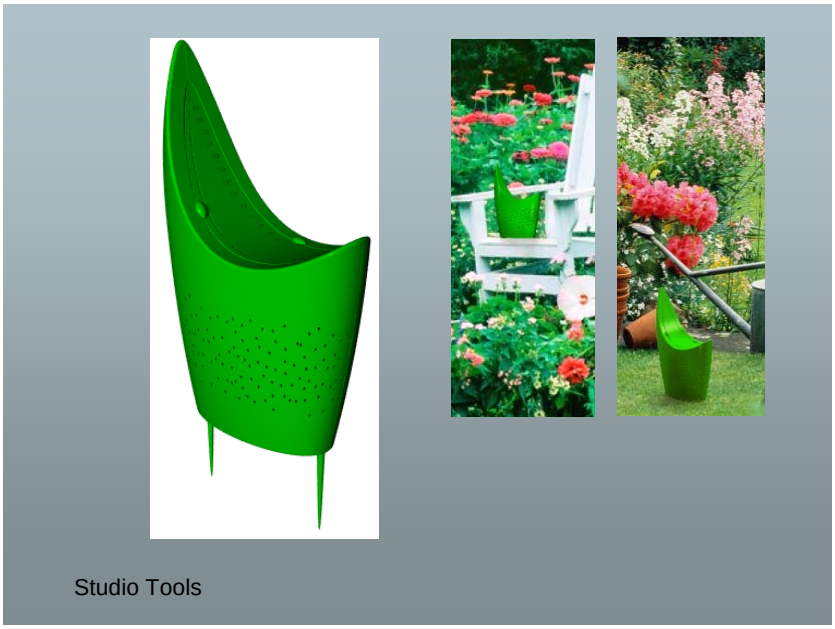


Figure 6b – Model in Studio Tools

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Reflexions on theory in practice

The semiotic method developed at the Department gives companies an effective tool for designing a product and its message. The method is applicable throughout the process of development. It can be used for establishing requirements of a product, for benchmarking of the semantic functions in one's own and competing products, and for evaluating different proposals of products during the entire development process.

But in order to utilise the method's full potential – be able to present the development of a product form as a process of building meaning – the practitioner must also have knowledge of semiotic theory and experience of how to employ Peirce's sign model. The interaction between user and product can then be further analysed and the discussion of particular expressions made more profound.

Practical application based upon theoretical understanding is thus important. At Chalmers the theoretical instruction in product semiotics is therefore strongly coupled to research in the field, and to practical application in projects of different kinds. The research and instruction are developed in symbiosis with each other.

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